

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_random_replay.
- Best TSPLIB holdout gap: phase6_random_replay.
- Best synthetic holdout gap: phase6_no_replay.

Run Metadata

- run_name: run_20260518_233254_d
- started_at_local: 2026-05-18 23:32:54
- finished_at_local: 2026-05-18 23:52:17
- duration_hhmm: 00:19
- duration_seconds: 1163.002
- seed_offset: 3000
- replicate_label: d
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.09339	0.0	0.05943	0.236772	0.115659	0.0	0.721692	0.76	0.029334
phase6_random_replay	random	score_only	False	0.083754	0.001628	0.05389	0.236772	0.135548	0.19457	0.749694	0.76	0.047462
phase6_failure_replay	failure	score_only	False	0.128074	0.0	0.081502	0.171206	0.288318	0.228311	0.704278	0.76	0.063433

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.108492	0.0	0.06904	0.236772	0.155909	0.19457	0.824501	0.76	0.035763
phase6_stratified_random_replay	stratified_random	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.221162	0.803495	0.56	-0.030996
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.130039	0.854129	0.56	-0.029158
phase6_residual_failure_replay	residual_failure	score_only	False	0.194753	0.0	0.123934	0.118662	0.278293	0.323448	0.0	0.76	0.0
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.08809	0.0	0.056057	0.171206	0.303258	0.172748	0.716584	0.76	0.049298
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.213387	0.064916	0.159398	0.171206	0.348017	0.17024	0.0	0.56	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.09339, synthetic 0.0, combined 0.05943.
- Accepted-epoch count 2, mean accepted code novelty 0.721692, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.115659, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency 0.029334 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.09339 across 7 instances; family means: ch=0.063484, kroD=0.077158, pcb=0.322502, pr=0.036113, rd=0.031732, st=0.059259.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 629, "cost": 947, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.505564, "residual_gap": 0.248012}, {"best_known_cost": 2579, "cost": 3778, "expected_gap": 0.400776, "family": "a", "name": "a280", "optimality_gap": 0.464909, "residual_gap": 0.064133},

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{"best_known_cost": 7542, "cost": 8832, "expected_gap": 0.245028, "family": "berlin", "name": "berlin52",  
"optimality_gap": 0.171042, "residual_gap": -0.073986}]
```

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.083754, synthetic 0.001628, combined 0.05389.
- Accepted-epoch count 3, mean accepted code novelty 0.749694, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.135548, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133496, mean selected residual gap -0.033587.
- Adaptation efficiency 0.047462 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.083754 across 7 instances; family means: ch=0.071883, kroD=0.030149, pcb=0.180137, pr=0.036113, rd=0.102781, st=0.093333.
- Panel synthetic_holdout mean gap 0.001628 across 4 instances; family means: clustered_gaussian=0.006511, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 7542, "cost": 10037, "expected_gap": 0.240918, "family": "berlin", "name": "berlin52", "optimality_gap": 0.330814, "residual_gap": 0.089896}, {"best_known_cost": 2579, "cost": 3416, "expected_gap": 0.400776, "family": "a", "name": "a280", "optimality_gap": 0.324544, "residual_gap": -0.076232}, {"best_known_cost": 629, "cost": 774, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.230525, "residual_gap": -0.025755}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.128074, synthetic 0.0, combined 0.081502.
- Accepted-epoch count 3, mean accepted code novelty 0.704278, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.288318, size bias 0.120322, and failure concentration 0.228311.
- Replay selection diversity 0.169313, mean selected expected gap 0.318675, mean selected residual gap -0.010989.
- Adaptation efficiency 0.063433 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 8, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.128074 across 7 instances; family means: ch=0.088632, kroD=0.084907, pcb=0.272579, pr=0.304949, rd=0.043489, st=0.013333.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 7542, "cost": 10015, "expected_gap": 0.236622, "family": "berlin", "name": "berlin52", "optimality_gap": 0.327897, "residual_gap": 0.091275}, {"best_known_cost": 2579, "cost": 3388, "expected_gap": 0.401939, "family": "a", "name": "a280", "optimality_gap": 0.313687, "residual_gap": -0.088252}, {"best_known_cost": 14379, "cost": 18452, "expected_gap": 0.322164, "family": "lin", "name": "lin105", "optimality_gap": 0.28326, "residual_gap": -0.038904}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.108492, synthetic 0.0, combined 0.06904.

- Accepted-epoch count 3, mean accepted code novelty 0.824501, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.155909, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132323, mean selected residual gap -0.014481.
- Adaptation efficiency 0.035763 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.108492 across 7 instances; family means: ch=0.100579, kroD=0.071898, pcb=0.26734, pr=0.103255, rd=0.041719, st=0.074074.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3468, "expected_gap": 0.403645, "family": "a", "name": "a280", "optimality_gap": 0.344707, "residual_gap": -0.058938}, {"best_known_cost": 7542, "cost": 7990, "expected_gap": 0.233148, "family": "berlin", "name": "berlin52", "optimality_gap": 0.059401, "residual_gap": -0.173747}, {"best_known_cost": 629, "cost": 663, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.054054, "residual_gap": -0.199364}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.803495, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.237974, mean selected residual gap 0.052751.
- Adaptation efficiency -0.030996 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.406514, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.005196}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.134817}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.328841, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": -2.8e-05}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.854129, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.130039.
- Replay selection diversity 0.266358, mean selected expected gap 0.168262, mean selected residual gap 0.025427.

- Adaptation efficiency -0.029158 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.406049, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.004731}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.137043}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.328006, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.000807}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.194753, synthetic 0.0, combined 0.123934.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive residual_archive with mean diversity 0.118662, mean hardness 0.278293, size bias 0.08672, and failure concentration 0.323448.
- Replay selection diversity 0.149216, mean selected expected gap 0.273635, mean selected residual gap 0.04886.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.194753 across 7 instances; family means: ch=0.103708, kroD=0.360618, pcb=0.22362, pr=0.340869, rd=0.205563, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19964, "expected_gap": 0.326657, "family": "lin", "name": "lin105", "optimality_gap": 0.388414, "residual_gap": 0.061757}, {"best_known_cost": 629, "cost": 815, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.295707, "residual_gap": 0.033704}, {"best_known_cost": 7542, "cost": 9666, "expected_gap": 0.233068, "family": "berlin", "name": "berlin52", "optimality_gap": 0.281623, "residual_gap": 0.048555}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.08809, synthetic 0.0, combined 0.056057.
- Accepted-epoch count 3, mean accepted code novelty 0.716584, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.303258, size bias 0.120322, and failure concentration 0.172748.
- Replay selection diversity 0.194842, mean selected expected gap 0.300271, mean selected residual gap 0.006324.
- Adaptation efficiency 0.049298 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.08809 across 7 instances; family means: ch=0.055342, kroD=0.080915, pcb=0.180866, pr=0.088028, rd=0.135398, st=0.020741.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3466, "expected_gap": 0.402482, "family": "a", "name": "a280", "optimality_gap": 0.343932, "residual_gap": -0.05855}, {"best_known_cost": 14379, "cost": 17624, "expected_gap": 0.326657, "family": "lin", "name": "lin105", "optimality_gap": 0.225676, "residual_gap": -0.100981}, {"best_known_cost": 7542, "cost": 9128, "expected_gap": 0.235826, "family": "berlin", "name": "berlin52", "optimality_gap": 0.210289, "residual_gap": -0.025537}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.17024.
- Replay selection diversity 0.199284, mean selected expected gap 0.301021, mean selected residual gap 0.048968.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.401086, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": 0.000232}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.127186}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.324988, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.003825}]

Judge Appendix

TSP Benchmark Suite Evaluation Summary

Overview

- **Best Conditions:**
- Held-out TSPLIB gap: phase6_random_replay (0.08375)
- Synthetic holdout gap: phase6_no_replay (0.0)
- Best transfer (combined evidence from TSPLIB and synthetic): phase6_random_replay (TSPLIB gap 0.08375, Transfer gap 0.05389)
- Total conditions: 9
- Replay Modes: none, random, failure, stratified_random, diversity_weighted, residual_failure, diversity_failure
- Selection Modes: score_only, novelty_gate

Transfer Performance (Held-out TSPLIB and Synthetic Holdouts)

Condition	TS PLI B G ap	Synth etic Gap	Tran sfer Gap	Replay Mode	Select ion Mode	Code Novelty (mean)	Archive Diversit y	Archive Hardne ss	Replay Failure Concentration
phase6_random_replay	0.0 837 5	0.001 628	0.053 89	random	score_ only	0.7497	0.2368	0.1355	0.195
phase6_no_replay	0.0 933 9	0.0	0.059 43	none	score_ only	0.7217	0.2368	0.1157	0.0
phase6_failure_replay	0.1 280 7	0.0	0.081 50	failure	score_ only	0.7043	0.1712	0.2883	0.228
phase6_random_replay_c ompression	0.1 084 9	0.0	0.069 04	random	novelt y_gate	0.8245	0.2368	0.1559	0.195
phase6_stratified_random _replay	0.2 133 9	0.064 92	0.159 40	stratified_ random	score_ only	0.8035	0.2368	0.2148	0.221
phase6_diversity_weighte d_replay	0.2 133 9	0.064 92	0.159 40	diversity_ weighted	score_ only	0.8541	0.2368	0.2148	0.130
phase6_residual_failure_r eplay	0.1 947 5	0.0	0.123 93	residual_f ailure	score_ only	0.0	0.1187	0.2783	0.323
phase6_diversity_failure_r eplay	0.0 880 9	0.0	0.056 06	diversity_ failure	score_ only	0.7166	0.1712	0.3033	0.173
phase6_diversity_failure_r eplay_compression	0.2 133 9	0.064 92	0.159 40	diversity_ failure	novelt y_gate	0.0	0.1712	0.3480	0.170

Interpretation

Transfer Metrics:

- The phase6_random_replay condition achieves the lowest (best) TSPLIB held-out gap (0.08375) and the best overall transfer gap (0.05389), making it the best performer for generalization to real-world instances.
- phase6_no_replay has perfect synthetic holdout performance but slightly worse TSPLIB gap; it is the best on synthetic probed instances but lacks replay.
- Conditions using failure-based or diversity-aware replay generally show higher TSPLIB gaps, indicating reduced transfer compared to random replay.
- Stratified random and diversity-weighted replay show poor transfer performance (TSPLIB gaps over 0.21 and transfer gaps ~0.16), indicating limited benefit to transfer.

- Residual failure replay conditions have moderate to poor transfer performance with failure concentration and archive hardness relatively high, but associated gaps are higher than the best random replay condition.

Replay Archive and Mechanism Metrics:

- **Archive Diversity:**

Highest diversity (~0.24) is with random and no-replay conditions; diversity failure and failure replay conditions show lower archive diversity (~0.17 or less), indicating less diverse failure cases are replayed.

- **Archive Hardness:**

Failure-based replay conditions (failure, residual failure, diversity failure) tend to have higher hardness (>0.28), suggesting that these archives contain harder failure cases.

- **Replay Failure Concentration:**

- Failure replay conditions exhibit moderate failure concentration (0.17 - 0.32), indicating replay focused on concentrated difficult cases.
- Random replay has lower concentration (~0.19 - 0.20).
- No replay condition naturally shows 0 failure concentration (0.0).

- **Selection Diversity:**

Random and diversity-weighted replay conditions have non-zero replay selection diversity (~0.19 - 0.26), while residual failure has lower diversity (~0.15).

Adaptation Efficiency:

- Highest positive adaptation efficiency is observed with phase6_failure_replay (0.0634) and phase6_diversity_failure_replay (0.0493), indicating better training adaptation.
- Random replay with score-only selection also adapts reasonably well (0.0475).
- Negative or zero adaptation efficiency observed in certain diversity-based replay modes with mixed transfer performance.

Code Novelty vs Transfer:

- Code novelty is generally high (0.7 - 0.85) across conditions using replay with random or diversity-based modes.
- Notably, phase6_residual_failure_replay and compression replay variations have zero mean code novelty.
- However, transfer performance does not always correlate positively with code novelty:
- E.g., phase6_residual_failure_replay has zero novelty but high transfer gap (~0.12 to 0.16).
- phase6_diversity_failure_replay_compression has zero novelty but high transfer gap (~0.16).
- This confirms that code novelty (lexical diversity) alone does not imply algorithmic improvement unless transfer metrics improve accordingly.

Detailed Notes on Key Conditions

phase6_random_replay

- Among the best in transfer metrics (lowest TSPLIB gap 0.08375; transfer gap 0.05389).

- Replay focused on random failure sampling, moderate failure concentration (0.19), good selection diversity (0.26), moderate archive hardness (0.1355).
- High code novelty (~0.75) but adaptation efficiency moderate (0.0475).
- Indicates replay diversity balanced with transfer generalization.

phase6_no_replay

- No replay; achieves best synthetic holdout gap (0.0).
- Slightly worse TSPLIB heldout gap (0.09339) vs random replay.
- Zero replay failure concentration and no replay selection diversity.
- Code novelty ~0.72, similar to random replay.
- Suggests that replay is not strictly required for synthetic holdout optimality, but random replay aids cross-domain transfer.

phase6_failure_replay

- Best adaptation efficiency (0.0634) but poorer transfer performance (TSPLIB gap ~0.128; transfer gap 0.0815).
- High archive hardness (0.29) and failure concentration (0.23).
- Replay focused on failure cases leads to more concentrated replay but generalizes less well.
- Code novelty lower (~0.70), with size bias in archive.

phase6_stratified_random_replay & diversity_weighted_replay

- High TSPLIB gaps (0.21+) and transfer gaps (~0.16), poor transfer.
- Replay failure concentration moderate (0.13-0.22).
- High archive descriptor diversity (~0.24) but lower adaptation efficiency (negative or near zero).
- Code novelty high (0.80+).
- Suggests complexity or selection mode may negatively impact transfer despite high novelty.

phase6_residual_failure_replay

- Zero code novelty, highest replay failure concentration (~0.32).
- Worse TSPLIB gap (0.19), moderate transfer gap (0.12).
- High archive hardness (~0.32) and size bias.
- Replay focuses on residual failures, highly concentrated but with limited adaptation and transfer.

phase6_diversity_failure_replay

- Moderate TSPLIB gap (0.08809) and transfer gap (0.056), among better transfer results.
- Good adaptation efficiency (0.0493), high archive hardness (0.31), moderate failure concentration (0.17-0.19).
- Code novelty moderate (0.72).
- Replay balances failure diversity and archive hardness better than failure replay alone.

phase6_diversity_failure_replay_compression

- Zero code novelty, high archive hardness (0.35), and failure concentration (~0.17).
- Poor transfer gaps (TSPLIB gap 0.21, transfer gap 0.16).

- Compression pressure reduces code novelty but transfer does not improve.

Failures and Size Bias

- Failure replay modes show increased archive size bias (>0.1) with more failures and residual failures.
- Random replay modes have negligible size bias (~ 0).
- Failure concentration is lowest with no replay (0.0), moderate in random replay (~ 0.19), and highest in residual failure replay (~ 0.32).
- Diversity-weighted replay shows moderate failure concentration and high complexity.

Summary

- **Best transfer performance is achieved by random replay (phase6_random_replay), combining a relatively low TSPLIB optimality gap (8.3%) and transfer gap (5.4%).**
- **No replay matches synthetic holdout but slightly underperforms random replay on TSPLIB transfer, indicating replay aids generalization beyond synthetic settings.**
- **Failure-based replay modes deliver higher adaptation efficiency but worse transfer and greater failure concentration, indicating overfitting to replayed failure cases may limit generalization.**
- **Archival diversity and size bias analyses show failure replay concentrates more on hard/difficult cases at the cost of replay diversity; random replay preserves a more diverse archive.**
- **Code novelty is high in random and diversity-based replay except in compression and residual failure modes where novelty drops to zero; however, increased code novelty does not always correspond to better transfer.**
- **Selection mode ('score_only' vs 'novelty_gate') impacts adaptation and code novelty but improvements in code novelty do not uniformly correspond to better transfer metrics.**

Recommendations

- Prioritize **phase6_random_replay** for transfer-based tasks where held-out TSPLIB optimality gap and transfer gap are critical.
- Employ replay strategies preserving **archive diversity** and avoiding excessive failure concentration **to favor generalization**.
- Avoid over-concentration on failure cases alone (residual_failure or failure replay) that hurt transfer despite good adaptation and high hardness.
- Code novelty metrics should be interpreted cautiously; algorithmic improvement must be backed by reduced optimality and transfer gaps, not lexical novelty alone.
- Compression strategies reduce code novelty but do not improve transfer; use with caution.

This conservative interpretation bases conclusions primarily on transfer (held-out TSPLIB and synthetic), with explicit attention to replay archive diversity, hardness, size bias, and failure concentration.